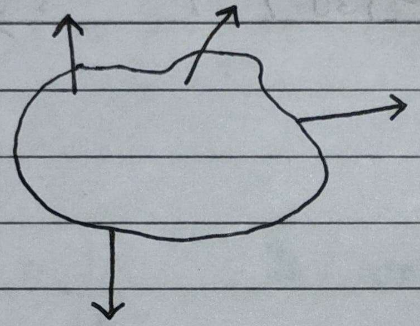
MODULE-I

Two Dimensional Non-concurrent Force System

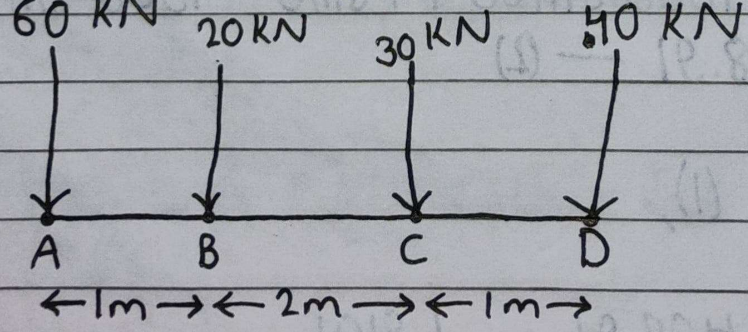
Non-concurrent Force System:

If All forces acting on a body within the same plane but their line of action do not intersect at a single point, then the system is Known as non-concurrent force system.



Resultant of Two Dimensional Co-planar Non-concurrent Force System

Q) Determine the resultant of four parallel forces acting on axle of a vehicle.



Sol^n:

$\Sigma F_x = 0$

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$$\Sigma F_y = -60 - 20 - 30 - 40 = 150 \text{ KN (Downward)}$$

$$\therefore R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

$$= \sqrt{0 + (-150)^2}$$

$$= 150 \text{ KN}.$$

Also,

$$\alpha = \tan^{-1}\left(\frac{\Sigma F_y}{\Sigma F_x}\right)$$

$$= \tan^{-1}\left(\frac{-150}{0}\right)$$

$$= \tan^{-1}(-\infty)$$

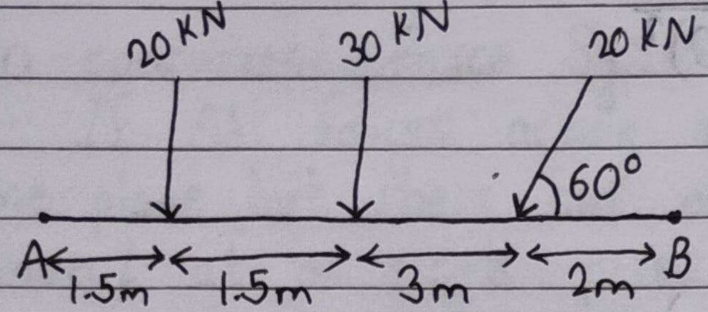
$$\therefore \alpha = -90^\circ \text{ (or } 270^\circ\text{)}.$$

$$\Sigma M_A = (60 \times 0) + (20 \times 1) + (30 \times 3) + (40 \times 4)$$
$$= 270 \text{ KNm}$$

$$\text{Distance } (x) = \frac{\Sigma M_A}{\Sigma F_y} = \frac{270}{150} = 1.8 \text{ m}.$$

Hence, the resultant is a force of 150 KN (downward) with an angle of -90° with horizontal and at a distance of 1.8 m to the right of point 'A'.

2). A system of loads acts on a beam AB as shown in the figure. Determine the resultant of the loads.



Sol^n:

$\Sigma F_x = -20 \cos 60 = -10$ kN

$\Sigma F_y = -20 - 30 - 20 \sin 60 = -67.32$ kN

Then,

$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2} = 68.058$ kN.

$\alpha = \tan^{-1}\left(\frac{\Sigma F_y}{\Sigma F_x}\right) = 81.55°$

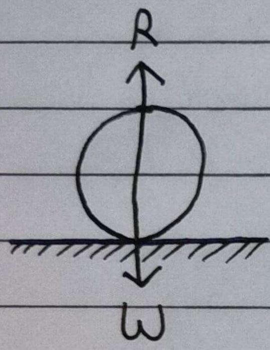
$\Sigma M_A = (20 \times 1.5) + (30 \times 3) + (20 \sin 60 \times 6)$
$= 223.92$ kNm

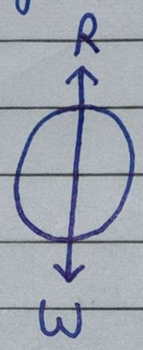
Distance, $x = \frac{\Sigma M_A}{\Sigma F_y} = 3.326$ m.

Free Body Diagram

Such a diagram of a body in which the body under consideration is freed from ~~the~~ all the contact surfaces and shows all the forces acting on it (Including reactions at contact surface), is called a free body diagram (FBD).

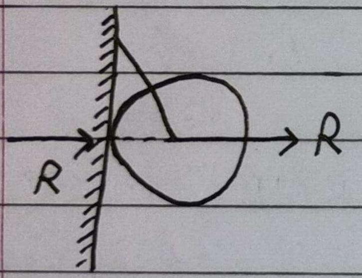
1.

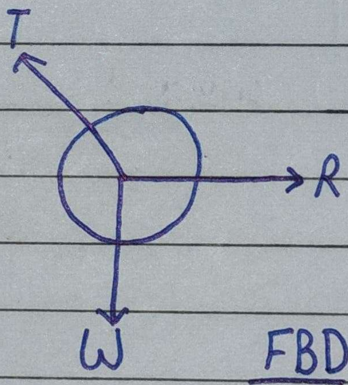




FBD

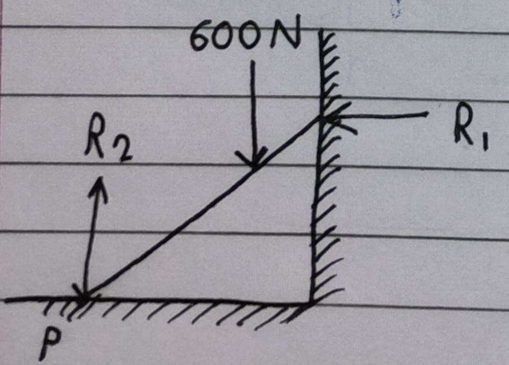
2.

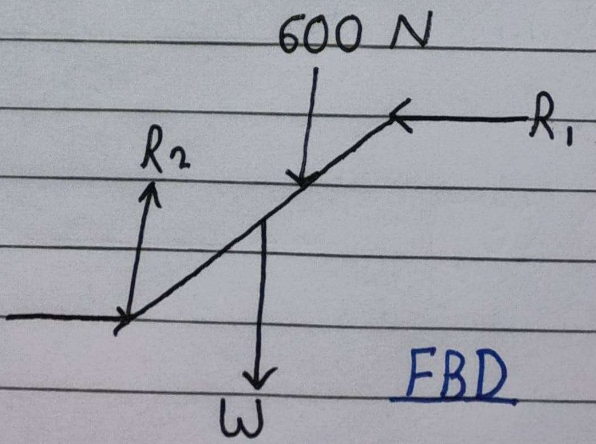




FBD

3.

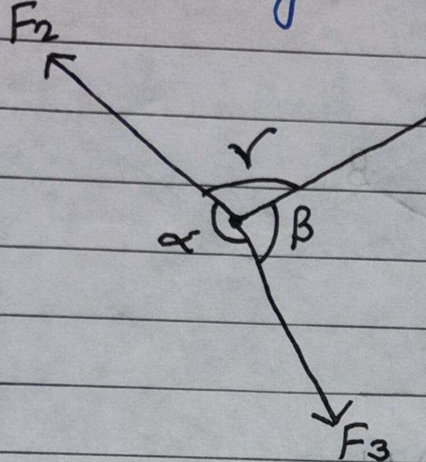




FBD

Lami's Theorem

If a body is in equilibrium under the action of three forces, each force is proportional to the sine of the angle between the other two forces.

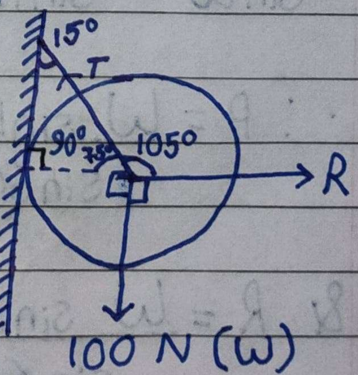


$$\boxed{\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}}$$

Numericals

Q) A sphere of weight 100 N is tied to a smooth wall by a string. Find the tension in the string and the reaction of the wall if the string makes an angle of 15° with the wall.

Sol^n:



From Lami's Theorem,

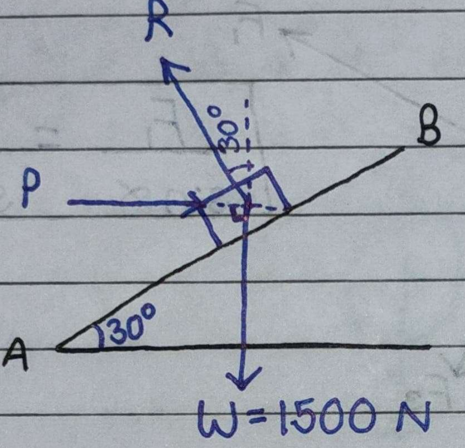
$$\frac{R}{\sin 165} = \frac{T}{\sin 90} = \frac{W}{\sin 105}$$

$$\therefore T = \frac{100 \sin 90}{\sin 105} = 103.527 \text{ N}.$$

$$\& \ R = \frac{100 \sin 165}{\sin 105} = 26.794 \text{ N}.$$

Q) Determine the horizontal force 'P' required to maintain equilibrium for a block of weight 1500 N, resting on a smooth plane inclined at 30° and find the normal reaction 'R' from the plane.

Solⁿ:



The angle between: W & P is 90°,
W & R is $180 + (-30) = 150°$
& P & R is $(90-30) = 60°$

From Lami's Theorem,

$$\frac{W}{\sin 60} = \frac{P}{\sin 150} = \frac{R}{\sin 90}$$

$$\therefore P = \frac{W \sin 150}{\sin 60} = \frac{1500 \times \sin 150}{\sin 60} = 866.0254 \text{ N.}$$

$$\& \ R = \frac{W \sin 90}{\sin 60} = \frac{1500 \times \sin 90}{\sin 60} = 1732.0508 \text{ N.}$$

Equilibrium of Connected Bodies

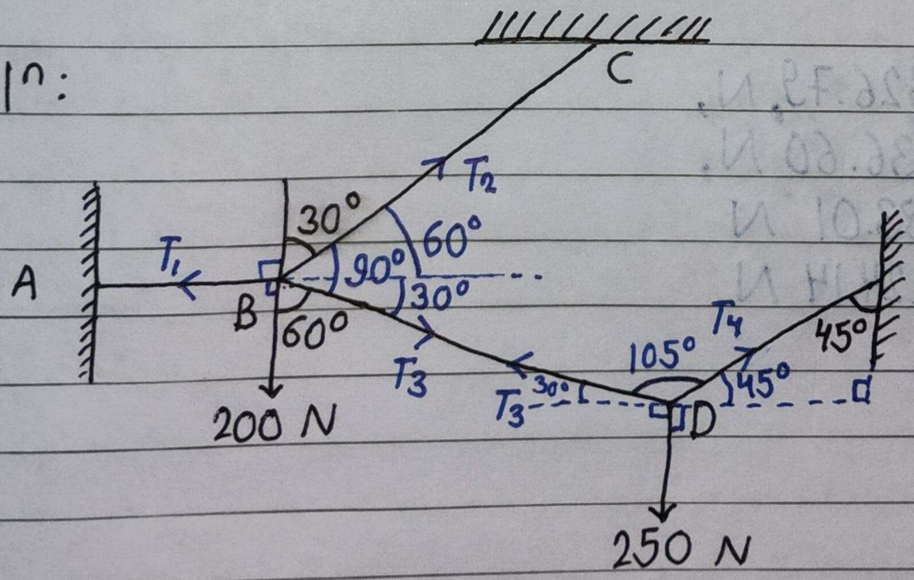
When two or more bodies are in contact with one another, the system of forces appears as it is a non-concurrent force system.

However, when each body is considered separately, in many situation, it turns out to be a set of concurrent force system. In such instances, first the body subjected to only two unknown forces, is to be analyzed followed by the analysis of other connected bodies.

Numericals

Q) A system of connected flexible cables is supporting two vertical forces 200 N and 250 N at point B and D respectively. Determine the forces in various segments of the cable?

Soln:



Applying Lami's Theorem at D,

$$\frac{T_3}{\sin 135} = \frac{T_4}{\sin 120} = \frac{250}{\sin 105}$$

$$\therefore T_3 = \frac{250 \times \sin 135}{\sin 105} = 183.01 \text{ N.}$$

$$\& \quad T_4 = \frac{250 \times \sin 120}{\sin 105} = 224.14 \text{ N.}$$

At point B,

$$-T_1 + T_3 \cos 30 + T_2 \cos 60 = 0 \quad \text{— (1)}$$

$$T_2 \sin 60 - 200 - T_3 \sin 30 = 0$$

$$\Rightarrow T_2 = \frac{200 + 183.01 \times \sin 30}{\sin 60} = 336.60 \text{ N.}$$

From (1),

$$T_1 = 183.01 \times \cos 30 + 336.60 \times \cos 60$$

$$\therefore T_1 = 326.7913 \text{ N.}$$

Hence,

$$T_1 = 326.79 \text{ N},$$

$$T_2 = 336.60 \text{ N},$$

$$T_3 = 183.01 \text{ N}$$

$$\& \quad T_4 = 224.14 \text{ N.}$$

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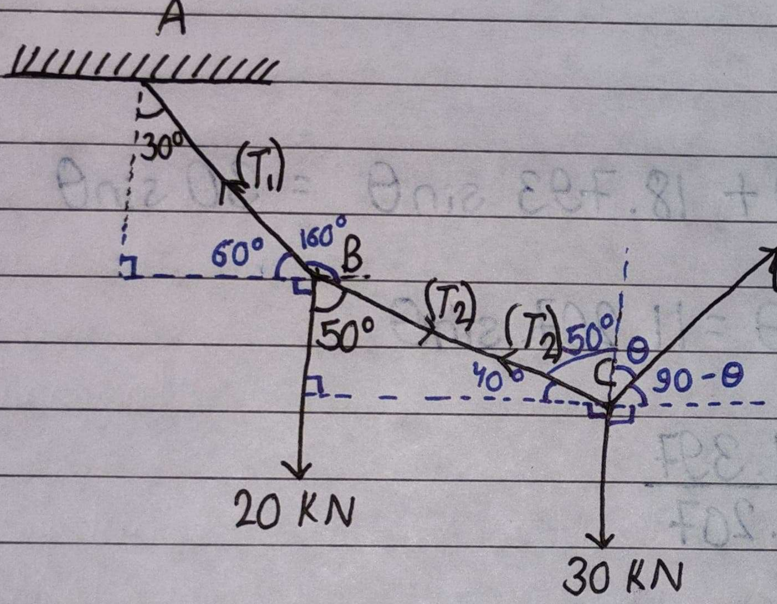
9) A wire rope is fixed at two point A and D. Two weights 20 KN and 30 KN are attached to it at B and C respectively. The weight rest with portion AB and BC inclined at angle 30° and 50° respectively, to the vertical. Find the tension in the wire in segments AB, BC and CD and also the inclination of the segment CD to vertical?



Sol^n:

Applying Lami's Theorem at point B,

$$\frac{T_1}{\sin 50} = \frac{T_2}{\sin 150} = \frac{20}{\sin 160}$$

$$\therefore T_1 = \frac{20 \times \sin 50}{\sin 160} = 44.795 \text{ KN}$$

$$T_2 = \frac{20 \times \sin 150}{\sin 160} = 29.238 \text{ KN}$$

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At C,

$$\frac{T_2}{\sin(180-\theta)} = \frac{T_3}{\sin 130} = \frac{30}{\sin(50+\theta)}$$

$$\Rightarrow 29.238 \sin(50+\theta) = 30 \sin(180-\theta)$$

$$\Rightarrow 29.238[\sin 50 . \cos\theta + \cos 50 . \sin\theta] = 30 . \sin\theta$$

$\cancel{\Rightarrow 29.238[}$

$$\Rightarrow 22.397 \cos\theta + 18.793 \sin\theta = 30 \sin\theta$$

$$\Rightarrow 22.397 . \cos\theta = 11.207 . \sin\theta$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{22.397}{11.207}$$

$$\Rightarrow \tan\theta = 1.998$$

$$\therefore \theta = 63.412^\circ .$$

$$T_3 = \frac{30 \times \sin 130}{\sin(50+63.412)} = 25.043 \text{ KN}.$$

Hence, the tension in the wire in segment AB is 44.795 KN, BC is 29.238 KN, CD is 25.043 KN and also the inclination of the segment CD to vertical is 63.412°.

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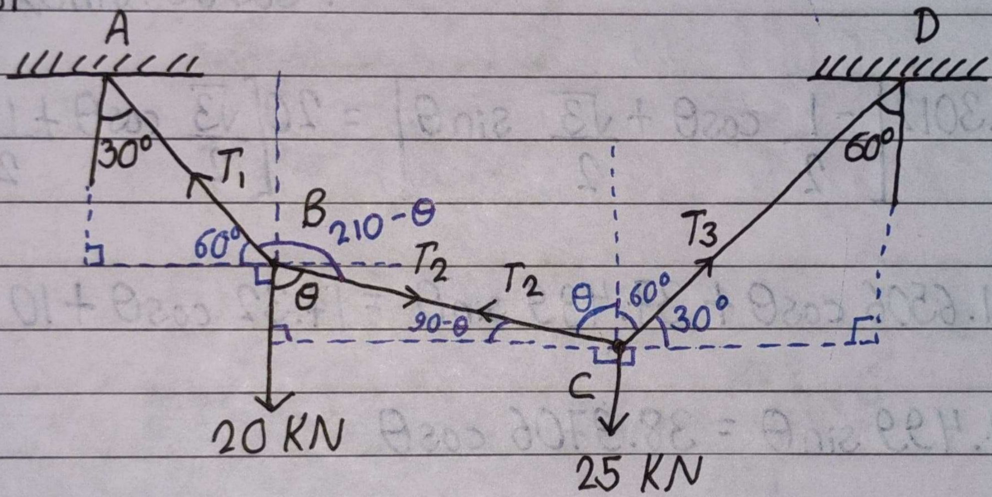
9) If a wire is fixed at two points A & D as shown in figure. Two weights 20 KN & 25 KN are supported at B & C, respectively. When equilibrium is reached, it is found that inclination of ~~AD~~ AB is 30° and that of CD is 60° to the vertical. Determine the tension in the segments AB, BC & CD of the rope and also the inclination of BC to the vertical?

Soln:



At point C,

$$\frac{T_2}{\sin 120} = \frac{T_3}{\sin(90+90-\theta)} = \frac{25}{\sin(60+\theta)}$$

$$\Rightarrow T_2 = \frac{25 \times \sin 120}{\sin(60+\theta)} \quad \text{--- (1)}$$

At point B,

$$\frac{T_1}{\sin\theta} = \frac{T_2}{\sin 150} = \frac{20}{\sin(210-\theta)}$$

$$\Rightarrow T_2 = \frac{20 \times \sin 150}{\sin(210-\theta)} \quad \text{—— (2)}$$

From (1) & (2),

$$25.\sin 120.\sin(210-\theta) = 20.\sin 150.\sin(60+\theta)$$

$$\Rightarrow 25 \times \frac{\sqrt{3}}{2} \times (\sin 210.\cos\theta - \cos 210.\sin\theta) = 20 \times \frac{1}{2} \times (\sin 60.\cos\theta + \cos 60.\sin\theta)$$

$$\Rightarrow 43.3012\left[-\frac{1}{2}\cos\theta + \frac{\sqrt{3}}{2}\sin\theta\right] = 20\left[\frac{\sqrt{3}}{2}\cos\theta + \frac{1}{2}\sin\theta\right]$$

$$\Rightarrow -21.6506\cos\theta + 37.499\sin\theta = 17.32\cos\theta + 10\sin\theta$$

$$\Rightarrow 27.499\sin\theta = 38.9706\cos\theta$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{38.9706}{27.499} \cancel{= 0.8123} = 1.417$$

$$\cancel{\Rightarrow \theta = \tan^{-1}(0.8123) = 39.08^\circ}$$

$$\therefore \theta = \tan^{-1}(1.417) = 54.788^\circ.$$

$$T_1 = \frac{20 \times \sin(54.788)}{\sin(210-54.788)} = 38.974 \text{ KN}.$$

$$T_2 = 23.851 \text{ KN}.$$

$$T_3 = 22.498 \text{ KN}.$$

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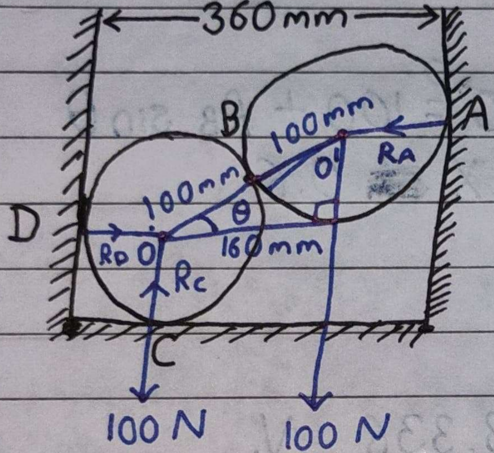
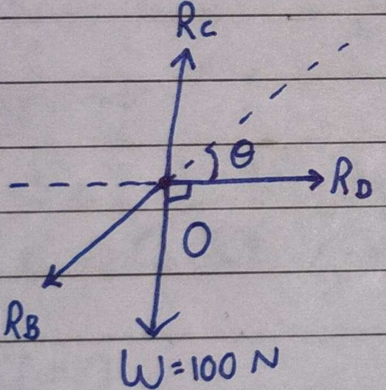
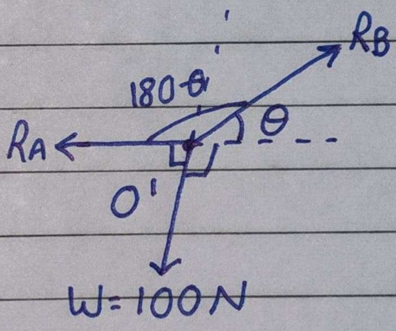
Q) Two smooth spheres each of radius 100 mm and weight 100 N, rest in a horizontal channel having vertical walls, the distance between wedges is 360 mm. Find the reactions at the point of contacts A, B, C & D.

Sol^n:







Now, $\cos\theta = \dfrac{160}{200} = 0.8 \Rightarrow \theta = 36.869°$

$\sin\theta = \sqrt{1-\cos^2\theta} = 0.6$

At O',

$$\frac{R_A}{\sin(90+\theta)} = \frac{R_B}{\sin 90} = \frac{100}{\sin(180-\theta)}$$

$$\Rightarrow R_A = \frac{100 \times \sin(90 + 36.869)}{\sin(180 - 36.869)} = 133.33 \text{ N}$$

$$R_B = \frac{100 \times \sin 90}{\sin(180 - 36.869)} = 166.67 \text{ N}$$

At O,

$$R_C \sin 90^\circ + \cancelto{0}{R_D \sin 0} = 100 + R_B \sin\theta$$

$$\Rightarrow R_C = 100 + 166.67 \times 0.6$$

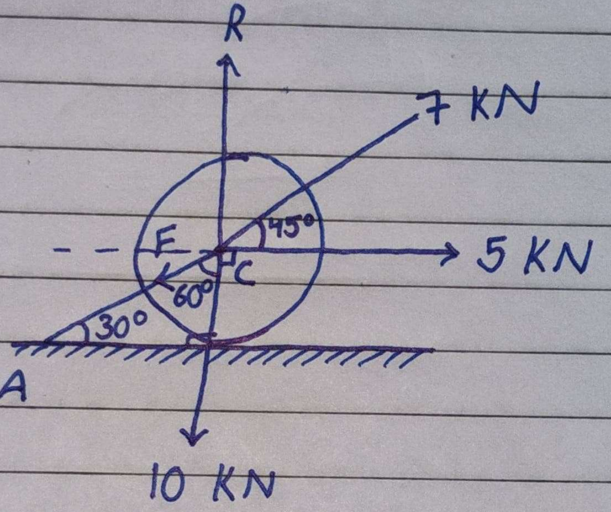
$$\therefore R_C = 200.002 \text{ N}$$

$$\& \; R_D = R_B \cos\theta = 133.336 \text{ N.}$$

Imp.

(9) A roller of weight 10 KN rest on a smooth horizontal plane (or floor) and is connected to the floor by the bar AC. Determine the force in the bar AC and reaction from the floor, if the roller is subjected to a horizontal force of 5 KN and inclined force of 7 KN as shown in figure.

Sol^n:



$\Sigma H = 0$

$\Rightarrow 5 - 7\cos 45 - F\cos 30 = 0$

$\Rightarrow F = 0.058$ KN.

$\Sigma V = 0$

$\Rightarrow R - 10 - 7\sin 45 - F\sin 30 = 0$

$\Rightarrow R = 14.978$ KN.